Keep the evidence, drop the person

Code-aware, privacy-first anonymization of doctor–patient encounters · Bosch @ NeuralWave 2026 · Team BlackPepperAI —Karapiperis Panagiotis Karapiperis Sotiris · repo: https://github.com/neural-wave/ch-blackpepper-ai

0.44 → 0.79
Held-out eval privacy, first pipeline → final, while code evidence kept rose **0.86 → 0.88**

0.971
Privacy score in the organizers' mid-review, on their own hidden metric

0 / 63
patient names recovered by a blind LLM attacker from our output (a standard de-id baseline leaks 5)

3.34 GB
peak VRAM end-to-end under a hard 4 GB cap, **7.9 s** per encounter, fully on-prem

1 · The idea: predict what to keep, not what to remove

The challenge has two goals that pull against each other: remove privacy-related information but keep diagnosis-supporting medical information. The first goal is evaluated by the percentage of removed privacy-related information while for the second goal we hope to keep the performance of an ICD code predictor on anonymized data the same as the performance using the original data. In addition, we measure the percentage of the kept related medical information. Our key observation is that the codes are given at inference, so the conflict disappears once medical information is split into two categories: 1) diagnosis-supporting information and 2) unrelated medical history/data. In total, we classify every information segment into three classes:

	what	treatment
a	personal: names, ages, dates, places, IDs, occupation and quasi-identifiers (family, hobbies, pets, trips, teams, small talk)	pseudonymize, generalize, remove
b	sensitive clinical data not supporting any code: unrelated history, drugs, procedures, labs	generalize to ICD block or remove
c	clinical evidence, cause or treatment of a coded diagnosis	keep verbatim (numbers ±3 %)

We aim to build a "**keep mask**" of code evidence and treat everything else as a privacy target to be removed. Here lies an important observation, medical data are personal data too: an unrelated HIV test, a past abortion or a psychiatric drug is exactly what a re-identification or discrimination attack would look for, so this information is removed even though no name is attached to it. Moreover, the context affects the class decision: *Parkinson* is a disease(c) or a surname(a), *golfing* is a hobby (a) unless the encounter suffers from "golfer's elbow" (c).

D2N001 · codes I11.0, I50.9, F32.A
hi , martha → **[PATIENT]** ...
martha is a 50-year-old → **[PATIENT]** is a **50s-year-old** female with ... congestive heart failure , depression and hypertension ...
i've been doing a lot of hiking . uh , went to washington last weekend to hike in northern cascades ... → **[small talk] [small talk]**
D2N004 · codes S39.012A, W18.49XA, Y93.53
james is a 57-year-old → **[PATIENT]** ... **50s-year-old**
uh , i was golfing and i hurt my back when i went for my backswing ← kept: coded cause

Real eval output. Orange = a (removed), green = c (kept verbatim).

Why privacy first

- Anonymized text is meant for **cloud LLMs**, which infer age, place and job from harmless-looking context (Staab et al., ICLR 2024). Names were solved early (NAME 0.99); the real leak was quasi-identifiers and unrelated diagnoses.
- In our benchmark one a/b occurrence weighs ~2.7× a c occurrence, so a keep rule only pays off above ~73 % precision. Being more careful about privacy brings better performance as measured by the metrics.
- The c-evidence is protected by construction, so privacy gains do not have to cost ICD utility. The challenging part is predicting between b and c.

2 · The pipeline

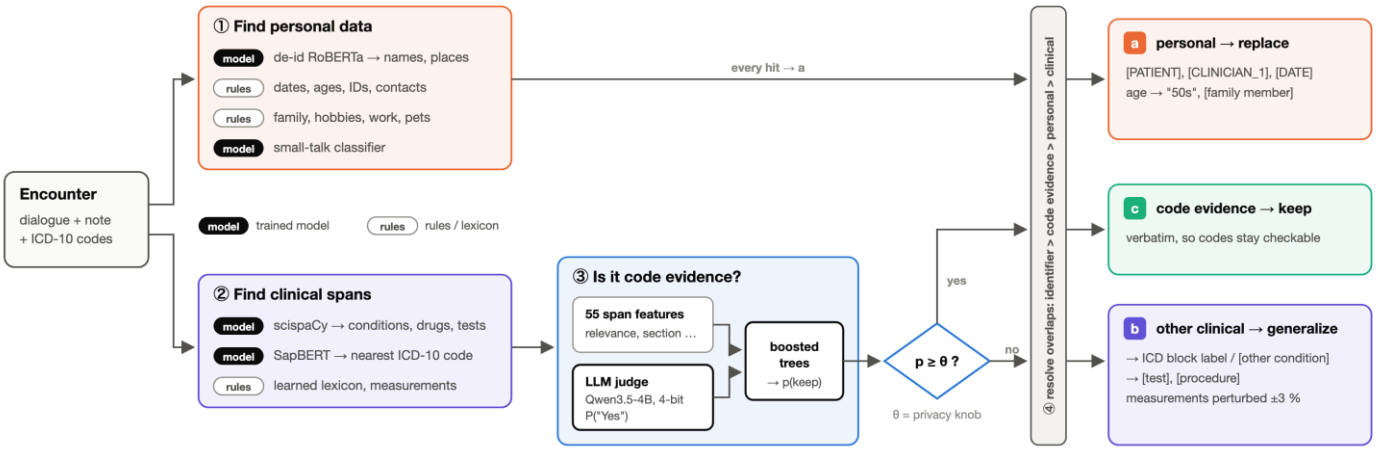


Figure 1. Two detection lanes (personal and clinical) feed a learned b/c decider; conflicts are resolved by priority, then every span is rewritten according to its class and logged for explainability.

Our pipeline is shown in Figure 1. We keep two separate tracks for personal and medical information. Personal information is identified using a heuristic rule-based system in conjunction with a pretrained de-identifier. The de-identifier receives enough context in sliding windows to disambiguate between similar terms belonging to different class (a,b, or c). We incrementally add more sophisticated systems such as a learnable "smallTalk" predictor. Everything that is detected as personal information is tagged for removal. Regarding the medical information, we are using a combination of rules and pretrained medical entity detecting models. Each such information is passed to the **decider** to decide between b) or c). Early prototypes used embedding similarity and thresholding however, as shown below we eventually hit a plateau. We converged to a learned classifier that is trained only on our pseudo-annotated data. The classifier receives handcrafted features such as the previously mentioned similarity in combination with logits from an LLM. Importantly, asking even 4B LLM to generate long form results, such as an anonymization tasks, is not viable under our given VRAM constraints. However, if we only ask for a binary answer and track the logits of "yes" and "no" it becomes feasible, allowing us to exploit the prior knowledge of these models. After the classifier decides between keeping and removing we apply a final filtering for overlaps and resolve conflicts. The final decisions are passed to the anonymization function that depending on the type of data performs replacing, generalization or perturbation. Summary:

model	role	why this one	size
obi/deid_roberta_i2b2	names, staff, places, dates	trained on clinical de-id (i2b2), far better on notes than general NER	355 M, fp16
scispaCy BC5CDR + en_core_sci_md	diseases, drugs, tests, procedures, findings tagging	CPU only; untyped entities catch what BC5CDR misses (labs, exams)	~120 MB ×2

model	role	why this one	size
SapBERT (PubMedBERT)	Span - code similarity, retrieval from 98k ICD-10-CM index	synonym-aware biomedical embeddings; 79 of 112 eval codes never occur in train, so retrieval beats a code classifier	110 M
Qwen3.5-4B, 4-bit NF4	zero-shot judge: “is this span evidence for one of these codes?”	one forward pass, P(Yes); won against MedGemma 1.5 4B and Qwen3.5-2B	3.3 GB peak
HistGradientBoosting	final keep/remove from 55 features + LLM score	tiny, trained on train out-of-fold; the threshold θ is the privacy knob	CPU
TF-IDF + logistic regression	small-talk units detection in the dialogue	cheap, trained on our own labels; removes chit-chat whole	CPU

3 · Our own benchmark: a pseudo ground truth

The organizers' metric is private, and early proxies (a general-purpose NER as “reference”, the pipeline's own scorer) were either noisy or circular. So we built a span-level ground truth for **all 207 encounters**:

- Written **annotation guidelines** (types per class, edge cases: eponyms, durations vs dates, injury causes are c, negated history is still b).
- Parallel **Claude agents** annotated the encounters in batches, given the codes; every span is an exact substring with type, code and a one-line reason; a validator rejects anything else.
- Result: train 1,536 a / 1,849 b / 7,251 c distinct strings, eval 729 / 1,192 / 3,411, with the same density per encounter on both splits.
- **Metric**: occurrence-weighted share of a and b strings that disappear (privacy) and of c strings that survive (c kept); score = mean of the two.
- **Protocol**: tune on train only; anything learned from labels runs **out-of-fold** (5 folds) on train; eval is scored once per final config. Every run lands in a leaderboard (46 configurations × 2 splits).
- **Process**: Every threshold selection and design choice is ablated and a global leaderboard of models is tracked.

4 · Results

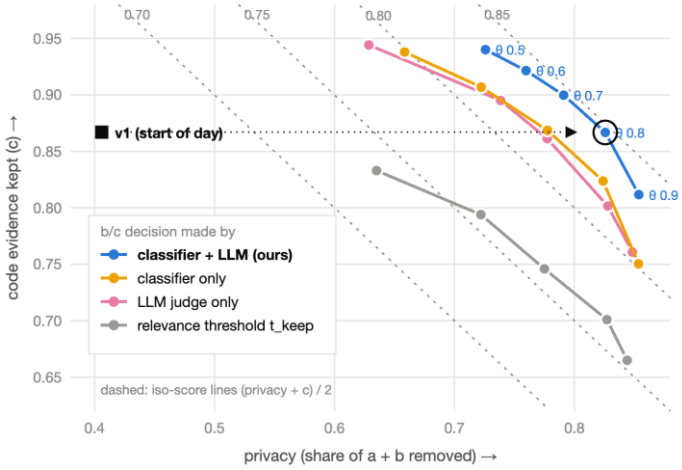


Figure 2. The privacy knob (train, out-of-fold). Each curve sweeps its decision threshold. The stacked decider dominates everything else; θ moves along it from utility-leaning (θ 0.5: c 0.94, privacy 0.73) to privacy-leaning (θ 0.9: privacy 0.85, c 0.81). We ship θ 0.8 (circled), the best score, leaning to privacy.

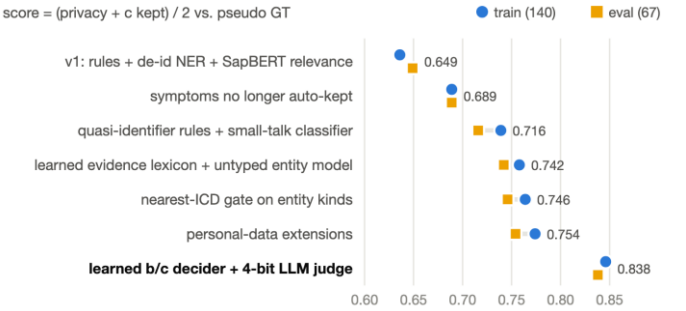


Figure 3. How the system evolved (one day, cumulative). Rules and lexicons bought +0.13; the learned b/c decider alone bought +0.07 (train) / +0.08 (eval). Train and eval stay close, so tuning did not overfit.

split	a removed	b removed	privacy	c kept	score
train · v1	0.593	0.232	0.406	0.867	0.636
train · final	0.895	0.761	0.826	0.867	0.846
eval · v1	0.636	0.313	0.440	0.858	0.649
eval · final	0.886	0.732	0.792	0.883	0.838

What the leaderboard taught us

- **Thresholds were a dead end.** We measured the precision of every keep rule: lexical match 0.99, same ICD category 0.99, learned lexicon 0.92, but same block 0.62–0.86 and same chapter 0.67. Eventually we hit a wall.
- **Classifier and LLM are complementary.** More advanced decision makers advance our performance. The LLM is the better ranker (eval AUC 0.915 vs 0.877) but too cautious; the classifier knows the data's habits (negation, note section, Assessment & Plan). Combining increase privacy-relate information removal while not decreasing important medical information.
- **Model choice matters more than size.** Qwen3.5-4B ≫ MedGemma 1.5 4B ≈ Qwen3.5-2B. MedGemma did not suite our specific use case, putting only 65 % of its next-token mass on Yes/No (Qwen 99 %),
- **With the LLM in, SapBERT features become redundant** for the decision (ties at 0.844); SapBERT stays for detection and for guarding places/activities that are coded causes.
- **A cascade** that only asks the LLM about uncertain spans keeps ~80 % of the gain with 18 % of the LLM calls — the efficiency option if latency matters.
- **Personal data:** 92 % of missed a-mentions were never detected at all, mostly quasi-identifiers and third-person phrasing in the note (“lives with wife”, “avid runner”). Rules for exactly those moved occupation 0.63 → 0.82 and quasi 0.66 → 0.74 at zero c cost.
- **Remaining gaps** (eval): unrelated procedures (0.42) and lab names (0.31) still leak; exam findings are the weakest kept evidence (0.81).

5 · Trade-offs, key decisions and what did not work

Decisions we would make again

- **Retrieval over classification** for code relevance: unseen codes dominate eval.
- **Graded actions**: unrelated diagnoses become their ICD block (“[influenza and pneumonia]”) instead of a blank when that is safe, which keeps the text readable for a downstream coder.
- **Consistent pseudonyms** per encounter ([PATIENT], [CLINICIAN_1]) and names found once are propagated to every mention in dialogue and note: an attacker cannot link an inconsistent replacement back.
- **Age as a decade** (“50s”, never “50–59”, which contains the exact age), holidays tied to an event as [DATE].
- **Measured everything**: each idea is an ablation switch, so we could drop the ones that did not pay.**I**

Ideas that did not work (or not as hoped)

- **Keeping all symptoms** as evidence (ICD chapter R) leaked: turning it off gave +0.13 privacy for −0.05 evidence kept.

- **Same-chapter / same-block relevance** kept unrelated diseases of the same organ system.
- **Untyped entity model, ungated**: +0.25 b removed but −0.21 c kept. Only a nearest-ICD similarity gate made it a net gain.
  Cross-field propagation of hobbies did nothing: the note paraphrases the dialogue, it does not repeat it.
- **A stricter injury cue** for keeping activities just swapped correct keeps for others.
- **MedGemma** as judge, and neighbor context for the small-talk classifier: no gain.
- **VRAM blow-up**: our first LLM runs peaked at 7–10 GB because full-vocabulary logits were kept for every position. Keeping only the last token and a hard CUDA cap brought it to 3.3 GB.

6 · Bonus/Additional challenges

Explainability. Every edit and every keep is written to `*.decisions.jsonl` with span, class, action, replacement, score, the matched code and a reason, e.g. *"code evidence for F32.A: similar to 'mood'"* or *"age generalized to a 10-year band"*. Coverage is 100 % by construction.

Efficiency. The CLI runs in phases so only one GPU model is resident at a time (detection models are released before the LLM loads). Measured end-to-end on eval under `--vram-cap-gb 4`: peak 3.34 GB, 7.9 s per encounter on an RTX 4090 at batch 1. Everything except the de-id NER, SapBERT and the judge runs on CPU; model outputs are cached on disk (warm runs ~3× faster), and the cascade cuts LLM calls by 82 %.

Without ICD codes. The pipeline can also run when no diagnosis codes are supplied. We tag the medical terms in the encounter, use SapBERT to retrieve the most relevant codes from the 98k ICD-10-CM index, and run the same anonymization logic on these predicted codes. Personal data removal and evidence retention stay the same, but b removal drops from 0.73 to 0.50. This is expected: the predicted codes are derived from the medical terms of the encounter itself, so most medical information now counts as evidence for some code and is kept.

eval vs. pseudo GT	a removed	b removed	c kept
codes given (default)	0.89	0.73	0.88
codes predicted (no input codes)	0.90	0.50	0.87

Defense on Privacy attacks. Threat model: we gave our anonymized output to a frontier cloud LLM (Claude), which is exactly where this data would end up. It acted as a blind attacker and tried to recover each patient's identity and personal details. Comparison: a standard de-identification baseline (removes only names, dates, ages, IDs and places) against our pipeline, on all 67 eval encounters. Direct identifiers: 0 of 63 names recovered (the baseline leaks 5). Age only comes back as the decade. Personal details (job, location, family, hobbies, life events): exact recovery falls from 81% with our initial baseline to 54% with our final model. Leftover personal mentions in the text: 42% with the baseline vs 11% with the final, consistent with the organizers' mid-review privacy score of 0.971. Re-identification: in a pool of only 67 patients, the attacker can still match most records. It does so through the clinical story we must keep for ICD coding (how the injury happened, smoking, pregnancy history). We cut confident correct matches from 72% to 48%. This is the privacy–utility trade-off made visible. What we learned: the attacker found concrete leaks, such as a clinic name and context that gives away a removed pharmacy name.

7 · Engineering

One package (`anonymizer/`) and one entry point (`python -m anonymizer.cli`); the output keeps the input schema and the evaluator asserts that `icd10_codes` pass through untouched. The pipeline is split into a model-heavy `prepare` and a replayable decision tail, so a θ sweep replays saved state instead of re-running models. Every idea is a config switch, every run is written to a JSON leaderboard, and a study log records why each default changed. Developed on a MacBook (MPS) and an RTX 4090 GPU box; the evaluation path is verified under the 4 GB cap.

8 · Use of AI tools

We used **Claude Code** (Anthropic) as a coding agent throughout: it wrote most of the code and experiment scripts under our direction, ran the sweeps and kept the study log; we set the strategy, the category scheme, the experiments to run and reviewed the results. The pseudo ground truth was annotated by **Claude agents** following our guidelines. This report was drafted with Claude Code from our study log and edited by us. At inference the only LLM is the locally run open-weight Qwen3.5-4B; no data leaves the machine.

Data & references. ACI-Bench encounters with ICD-10 annotations from Yuan, Moy et al., EMNLP Industry 2025 (CC BY-NC 4.0). Liu et al., SapBERT, NAACL 2021 · Neumann et al., scispaCy, BioNLP 2019 · Staab et al., Beyond Memorization, ICLR 2024 · Pilán et al., Text Anonymization Benchmark, 2022 · Sweeney, k-anonymity, 2002.